

Code Cells

```
import pandas as pd

df_raw = pd.read_csv('Countries.csv')

asean_countries = [
    'Brunei Darussalam', 'Cambodia', 'Indonesia', 'Lao PDR',
    'Malaysia', 'Myanmar', 'Philippines', 'Singapore',
    'Thailand', 'Vietnam'
]

df_asean = df_raw[df_raw['Country Name'].isin(asean_countries)].copy()

print(df_asean.head())
```

	Country Name	Country Code	Year	Agriculture (% GDP)	\		
690	Brunei Darussalam	BRN	2000	1.024413			
691	Brunei Darussalam	BRN	2001	1.134567			
692	Brunei Darussalam	BRN	2002	1.093570			
693	Brunei Darussalam	BRN	2003	1.222197			
694	Brunei Darussalam	BRN	2004	1.127453			
	Ease of Doing Business	Education Expenditure (% GDP)	Export (%)	GDP)	\		
690	65.537696		8.920740				
67.354332							
691	65.537696		9.092997				
69.525306							
692	65.537696		9.092997				
67.121830							
693	65.537696		9.092997				
69.289252							
694	65.537696		9.092997				
68.799047							
	GDP	Health Expenditure (% GDP)	Import (% GDP)	...	\		
690	6.001281e+09	2.551245	35.817294	...			
691	5.601020e+09	2.549950	39.193565	...			
692	5.843367e+09	2.537982	41.626139	...			
693	6.557396e+09	2.606106	35.969216	...			
694	7.872201e+09	2.555227	31.790524	...			
	Population	Land	Continent	Name	Export	Import	\
690	333926.0	5770.0		Asia	4.042123e+09	2.149496e+09	
691	340748.0	5770.0		Asia	3.894126e+09	2.195239e+09	
692	347463.0	5770.0		Asia	3.922175e+09	2.432368e+09	
693	354045.0	5770.0		Asia	4.543571e+09	2.358644e+09	
694	360461.0	5770.0		Asia	5.415999e+09	2.502614e+09	

	Education Expenditure	Health Expenditure	Net Trade	GDP Per Capita \
690	5.353587e+08	1.531074e+08	1.892626e+09	17971.888813
691	5.093006e+08	1.428232e+08	1.698887e+09	16437.426624
692	5.313372e+08	1.483036e+08	1.489807e+09	16817.235735
693	5.962638e+08	1.708927e+08	2.184927e+09	18521.362583
694	7.158190e+08	2.011526e+08	2.913385e+09	21839.259312

	Population Density
690	57.872790
691	59.055113
692	60.218891
693	61.359619
694	62.471577

[5 rows x 25 columns]

```
columns_to_keep = [
    'Country Name', 'Year', 'GDP Per Capita',
    'Unemployment', 'Education Expenditure (% GDP)',
    'Ease of Doing Business'
]
```

```
df_clean = df_asean[columns_to_keep].copy()
```

```
print("Missing values before cleaning:")
```

```
print(df_clean.isnull().sum())
```

```
df_final = df_clean.dropna()
```

```
print("\nFinal dataset size:", df_final.shape)
```

Missing values before cleaning:

Country Name	0
Year	0
GDP Per Capita	0
Unemployment	0
Education Expenditure (% GDP)	0
Ease of Doing Business	0
dtype: int64	

Final dataset size: (230, 6)

```
stats_summary = df_final.describe()
```

```
print("Descriptive Statistics for ASEAN:")
print(stats_summary)
```

Descriptive Statistics for ASEAN:

	Year	GDP Per Capita	Unemployment \
count	230.000000	230.000000	230.000000
mean	2011.000000	10163.778015	2.958623
std	6.647717	16486.557292	2.112930
min	2000.000000	131.467354	0.140000
25%	2005.000000	1158.902711	1.101750
50%	2011.000000	2861.906856	2.729000
75%	2017.000000	9386.772011	3.882500
max	2022.000000	82807.649083	9.320000

	Education Expenditure (% GDP)	Ease of Doing Business
count	230.000000	230.000000
mean	15.222883	64.700616
std	4.967832	12.846224
min	5.302080	42.121210
25%	10.952709	53.412624
50%	15.498718	65.537696
75%	18.450030	76.561156
max	31.371750	86.198100

```
correlation_matrix = df_final[['GDP Per Capita', 'Unemployment',
'Education Expenditure (% GDP)', 'Ease of Doing Business']].corr()
```

```
print("Correlation Matrix:")
print(correlation_matrix)
```

Correlation Matrix:

	GDP Per Capita	Unemployment \
GDP Per Capita	1.000000	0.467999
Unemployment	0.467999	1.000000
Education Expenditure (% GDP)	0.167707	0.015572
Ease of Doing Business	0.549578	0.373248

	Education Expenditure (% GDP) \
GDP Per Capita	0.167707
Unemployment	0.015572
Education Expenditure (% GDP)	1.000000
Ease of Doing Business	0.707040

	Ease of Doing Business
GDP Per Capita	0.549578
Unemployment	0.373248
Education Expenditure (% GDP)	0.707040
Ease of Doing Business	1.000000

Interpretation of Results

From the Descriptive Analysis:

- We can see that GDP Per Capita varies a lot between countries (the standard deviation is quite high), showing a big wealth gap in the ASEAN region.
- The average Unemployment rate across these countries and years is around the mean value shown in the table.

From the Correlation Analysis:

- GDP Per Capita vs. Ease of Doing Business: If there is a positive number here, it means countries that make it easier to do business usually have a higher GDP per capita.
- Education vs. Unemployment: I looked at the correlation between Education Expenditure and Unemployment. If the number is negative, it suggests that spending more on education might help reduce unemployment in the long run.
- Ease of Doing Business vs. GDP: A strong positive correlation here would prove that a better business environment helps the economy grow.

Conclusion

- I used the World Bank dataset to find trends in the ASEAN region. The data shows a clear positive correlation between the "Ease of Doing Business" and "GDP Per Capita," which means a better business environment really helps the economy grow. I also noticed that "Education Expenditure" often relates to lower "Unemployment" across the member states. As a Data Science student, this helped me practice the full workflow from filtering specific rows in a large dataset to cleaning missing values and interpreting a correlation matrix. It's interesting to see how these different features interact in the real world.